



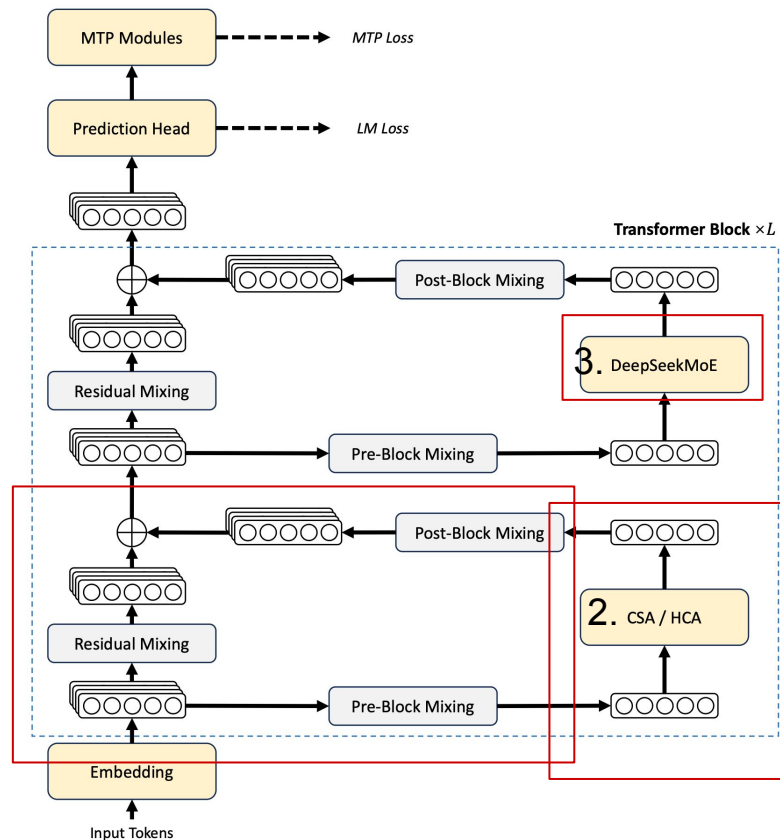
DeepSeek V4 in vLLM

The model, the performance, and the next steps

Recap on Model Architecture

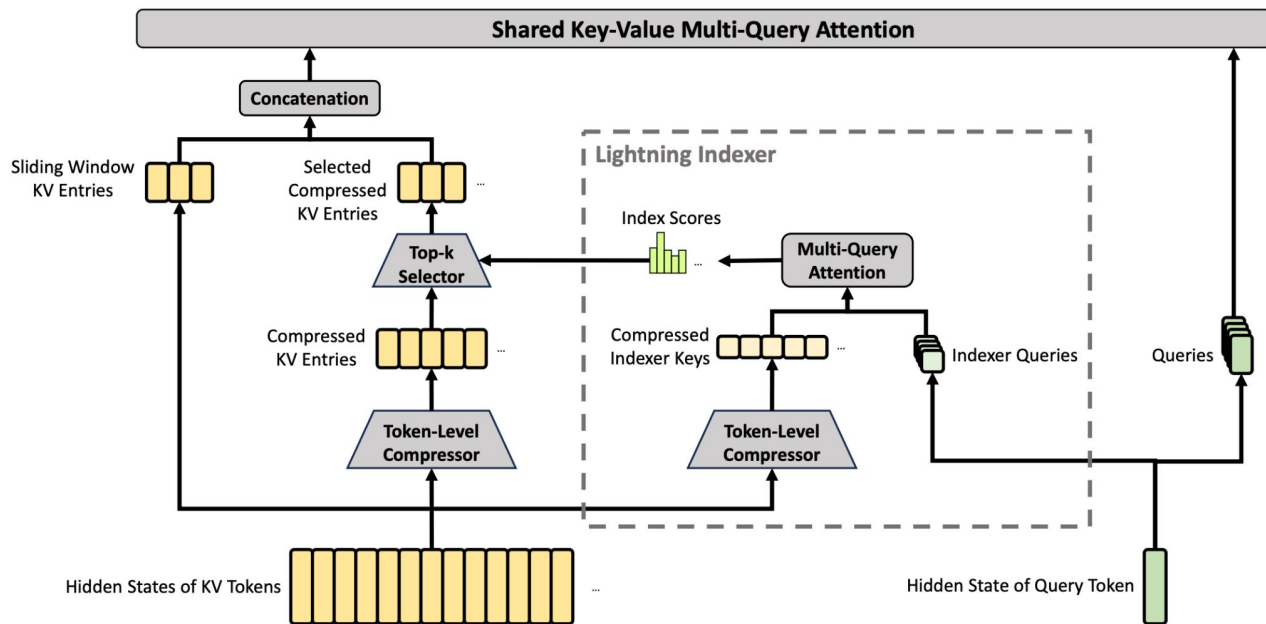
1. mHC
2. Compressed sparse attention
3. Highly compressed attention
4. MoE
 - a. Hash routing
 - b. MegaMoE

1. mHC

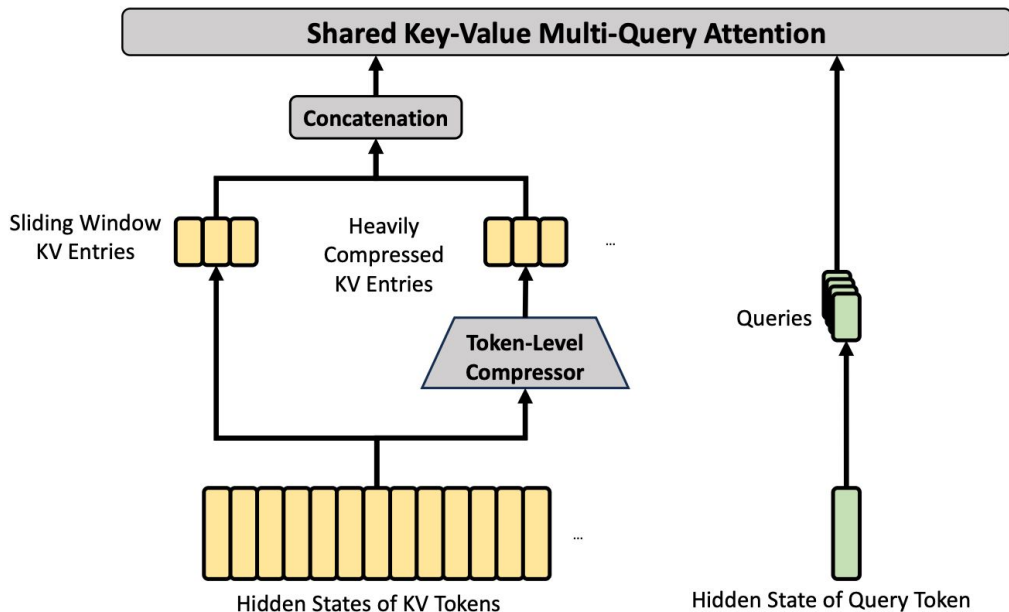


Recap on Model Architecture: CSA

https://vllm.ai/blog-assets/interactive_pages/c4a.html



Recap on Model Architecture: HCA



KV Cache Management: Overview

C4A (CSA): compressed sparse attention

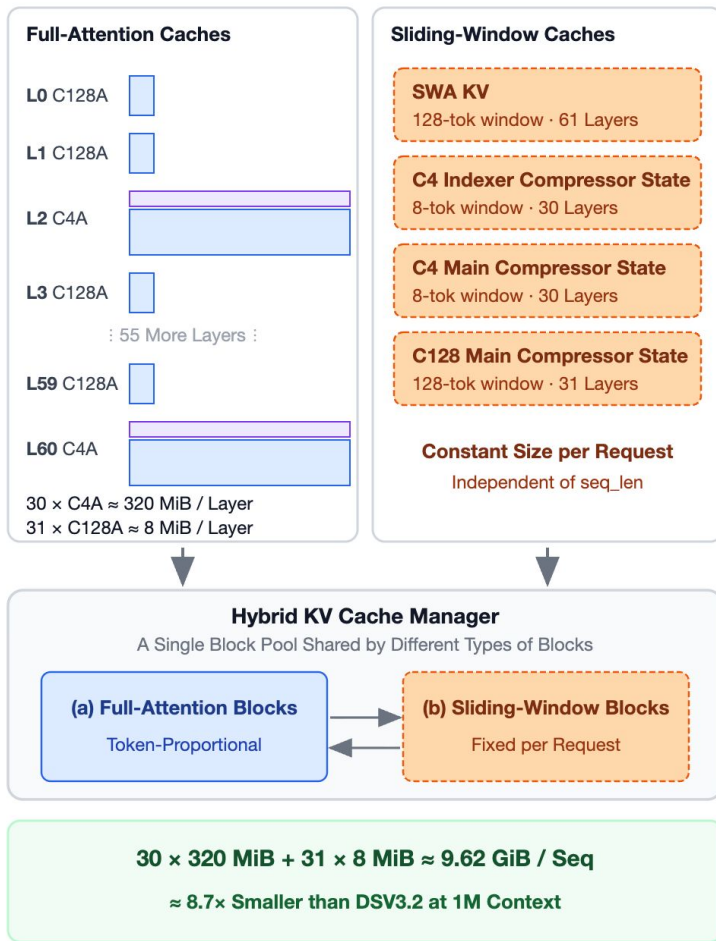
C128A (HCA): highly compressed attention

The hidden (yet simple) math behind the numbers:

- Assuming **block_size=256**
 - C4: 64 entries per block
 - C128: 2 entries per block
 - Referred to as **storage_block_size** in code
- C4 indexer (**head_dim=128**):
 - FP16: $64 \times 128 \times 2B = 16384$ B/block, or 64 B/token
 - FP8**: $64 \times 132B = 8448$ B/block, or 33 B/token
 - FP4: $64 \times 68B \dots$
- C4 attention (**head_dim=512**):
 - FP16: $64 \times 512 \times 2B = 65536$ B/block, or 256 B/token
 - FP8**: $64 \times 584B = 37376$ B/block, or 146 B/token
- C128 attention (**head_dim=512**):
 - FP16: $2 \times 512 \times 2B = 2048$ B/block, or 8 B/token
 - FP8**: $2 \times 584B = 1168$ B/block, or ~4.6 B/token

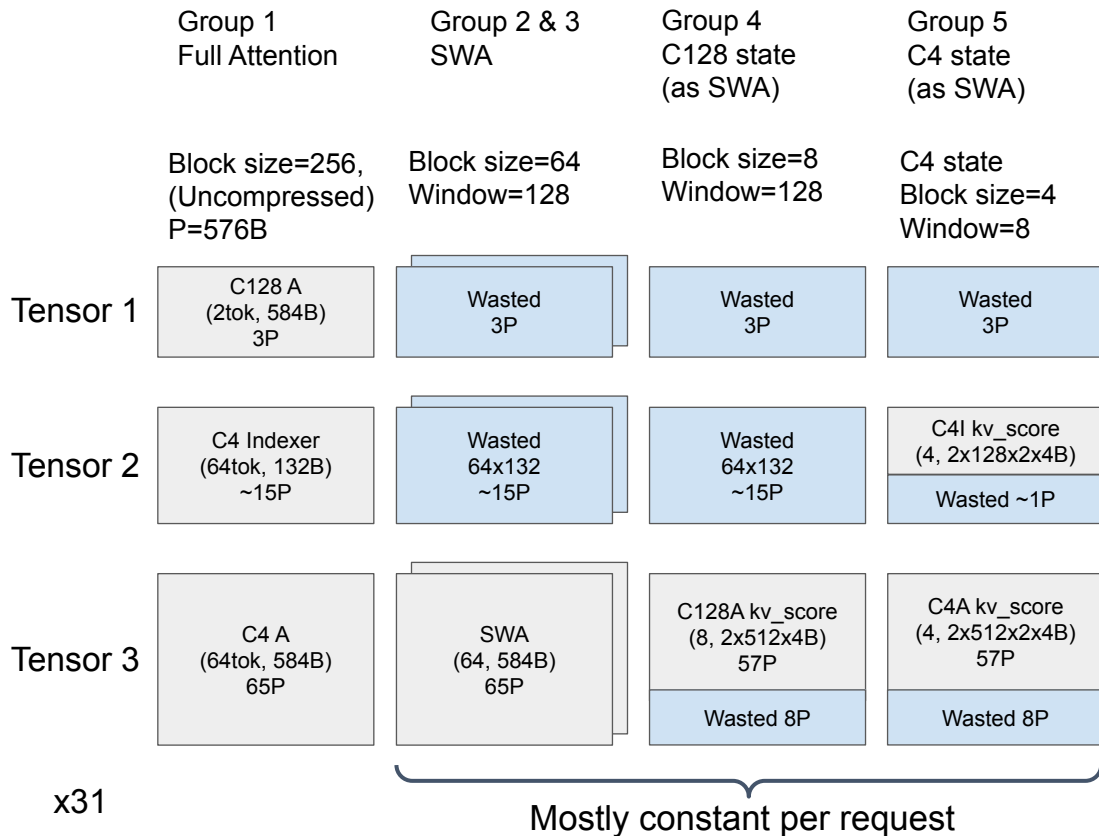
DeepSeek V4 Hybrid KV Cache

Indexer Main MLA Sliding Window



KV Cache Management: Flexible Memory Sharing

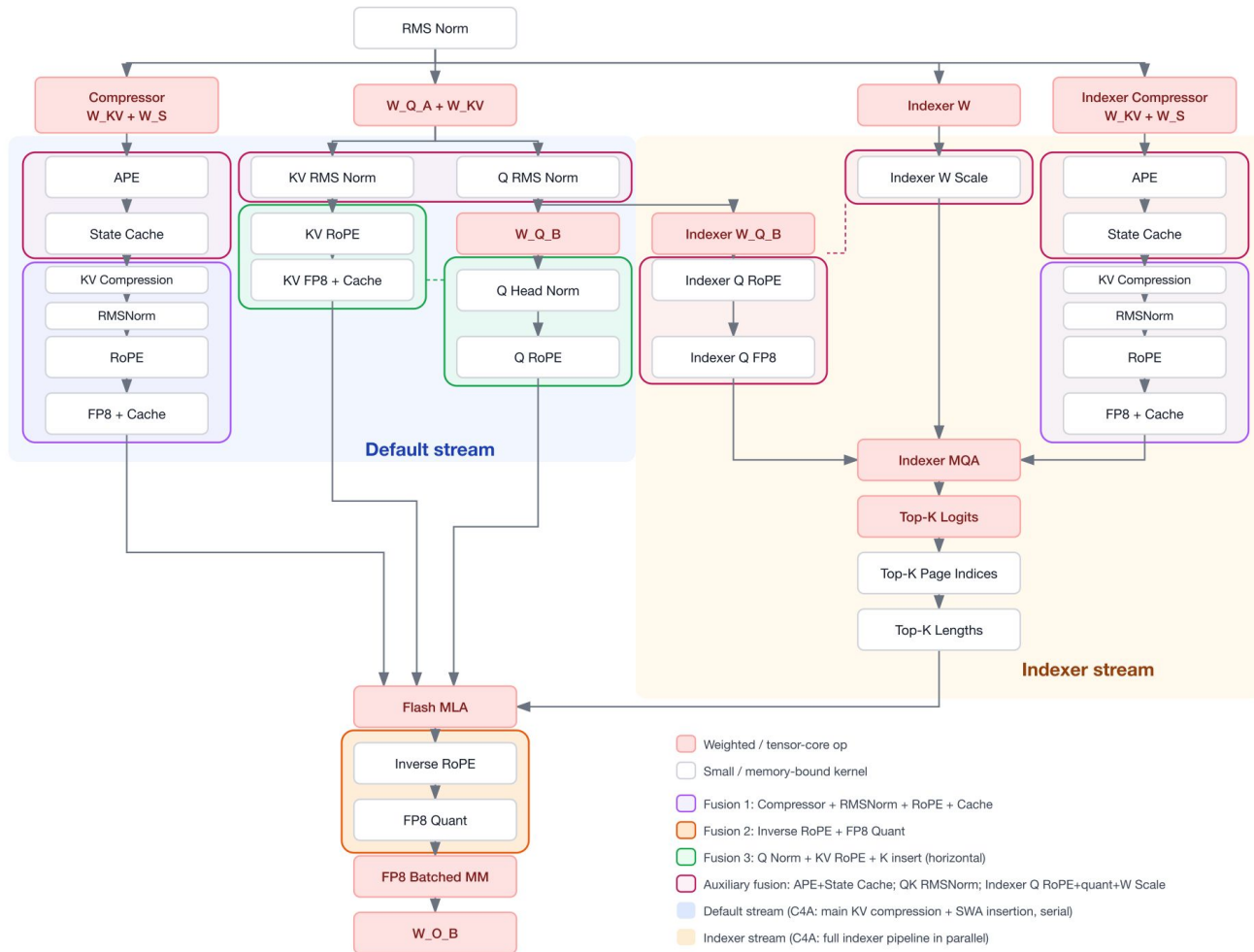
- Uniform logical block size
 - Defined on native tokens, fixed to be 256
 - Basic unit for scheduling and cache hit
- Compressor state as SWA
 - C4: window_size=8
 - C128: window_size=128



Kernel Optimizations

1. Kernel fusion
2. Multi-stream

DeepSeek V4 - C4A Decode Path



Key Source Code Files

- Entry + model definition: `models/deepseek_v4.py`
- High-level attention layer: `layers/deepseek_v4_attention.py`
 - Including the indexer and the main attention module
- Compressor (for both CSA and HCA): `layers/deepseek_compressor.py`
 - Including the compressor state cache and metadata builder
- SWA KV cache: `mha/sparse_mha.py`
 - Including the SWA cache and metadata builder
- MegaMoE: inlined in `models/deepseek_v4.py`
- KV cache layout: `core/kv_cache_utils.py:_get_kv_cache_config_deepseek_v4()`



THANK YOU

DOCS

docs.vllm.ai

GITHUB

github.com/vllm-project/vllm

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